Weekly Project Report – Week 2

Unified Generative AI Models via Command-Line Interface

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Weekly Summary

In Week 2 of our project, our main focus was on implementing key components of our unified generative AI system through a standardized Command-Line Interface (CLI). The objectives included:

- Generating images using a GAN model,
- Fine-tuning the GPT-2 model with a custom dataset to generate text,
- Ensuring all components are CLI-compatible for seamless integration.

Achievements

- We successfully fine-tuned the DistilGPT-2 model using our own dataset.
- Due to limited hardware access, training was conducted using CPU-only PyTorch implementation.
- Post-training, the model was able to generate coherent and contextually relevant text outputs based on provided prompts, confirming that the base functionality is operational.
- Successfully trained the GAN model and generated sample images to evaluate its performance.
- Developed a command-line interface (CLI) for the GAN model to streamline training and image generation tasks.
- Ensured the CLI is compatible with the project's overall structure and user workflow.

Challenges and Improvements

Although the results met our baseline expectations, some challenges were noted:

- Limited hardware led to slower training times and restricted model depth, with dependencies conflict of models.
- A few inconsistencies in output indicate the need for further refinement.

To address these, we plan to:

- Run additional training epochs to improve accuracy and coherence.
- Explore optimization techniques to enhance performance under CPU constraints.